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Лабораторная работа №1
«Intro Exp»
по дисциплине «Операционные системы»

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1. Задание

Introductory Experiment: Экспериментальное введение в курс «Операционные системы».

Экспериментально сравнить время работы выданных прототипов приложений, моделирующих отдельные сценарии нагрузки, характерные при работе с данными: чтение/обновление записей на диске в последовательном/случайном порядке.
Вариант «Read vs Write,Cache,10M,Seq».

2. Подготовка окружения

Для корректности эксперимента необходимо настроить чистое окружение.
В качестве стенда выбрана виртуальная машина с ОС Debian 13, сконфигурированная следующим образом:

- выделено 4 vCPU из 16 vCPU хоста
- ядра хоста закреплены за ядрами ВМ
- кэширование диска в гипервизоре отключено
- взаимодействие с сервером происходит по ssh

Фоновые процессы:

```
user@debian:~/lab1$ top -bn1 | head -20
top - 22:02:38 up  1:01,  1 user,  load average: 0.01, 0.02, 0.00
Tasks: 123 total,   1 running, 122 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3921.3 total,   3587.1 free,    347.8 used,    201.5 buff/cache
MiB Swap:   1101.0 total,   1101.0 free,      0.0 used.   3573.5 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   23820  14588  10768 S   0.0   0.4   0:01.00 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tas+
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tas+
     16 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tas+
```

Частота процессора хоста установлена на perfomance:

```
❯ cpupower frequency-info
analyzing CPU 11:
  driver: amd-pstate-epp
  CPUs which run at the same hardware frequency: 11
  CPUs which need to have their frequency coordinated by software: 11
  energy performance preference: performance
  hardware limits: 411 MHz - 4.55 GHz
```

Турбобуст не контролировался.

Прерывания обрабатываются vCPU 2 и 3 (это одно физическое ядро), следовательно запускать исследуемую программу нужно на vCPU 0 и 1:

Average:	CPU	%usr	%nice	%sys	%iowait	%irq	%soft	%steal	%guest	%gnice	%idle
Average:	all	0.01	0.00	0.05	0.00	0.00	0.01	0.02	0.00	0.00	99.90
Average:	0	0.03	0.00	0.03	0.00	0.00	0.00	0.00	0.00	0.00	99.95
Average:	1	0.00	0.00	0.05	0.00	0.00	0.00	0.00	0.00	0.00	99.95
Average:	2	0.03	0.00	0.03	0.00	0.00	0.03	0.08	0.00	0.00	99.84
Average:	3	0.00	0.00	0.11	0.00	0.00	0.03	0.00	0.00	0.00	99.86

3. Подготовка данных

Я сгенерировал два файла:

```
> python3 ../os-course/lab/util/graphgen.py -s 10M --seed 471855 --topology chain -b 0.5 --min-step-pages 2 -o graph-rand.bin
> python3 ../os-course/lab/util/graphgen.py -s 10M --seed 471855 --topology sequential -o graph-seq.bin
```

и скомпилировал тестируемые программы для обхода.

4. Исследование graph_traverse

4.1. Паспорт системы

См. приложение 1

4.2. Baseline

```
Iteration 1/1 (read): traversing graph-rand.bin ... OK (436905 nodes processed)

real    0m0.812s
user    0m0.142s
sys     0m0.533s
Iteration 1/1 (read): traversing graph-seq.bin ... OK (436905 nodes processed)

real    0m0.555s
user    0m0.176s
sys     0m0.379s
Iteration 1/1 (read): traversing graph-rand.bin ... OK (436905 nodes processed)

real    0m21.166s
user    0m0.175s
sys     0m4.813s
Iteration 1/1 (read): traversing graph-seq.bin ... OK (436905 nodes processed)

real    0m16.529s
user    0m0.231s
sys     0m3.891s
Iteration 1/1 (write): traversing graph-rand.bin ... OK (436905 nodes processed)

real    0m0.707s
user    0m0.137s
sys     0m0.570s
Iteration 1/1 (write): traversing graph-seq.bin ... OK (436905 nodes processed)

real    0m0.665s
user    0m0.168s
sys     0m0.496s
Iteration 1/1 (write): traversing graph-rand.bin ... OK (436905 nodes processed)
```

```
real    0m59.437s
user    0m0.518s
sys     0m12.372s
Iteration 1/1 (write): traversing graph-seq.bin ... OK (436905 nodes processed)

real    1m4.513s
user    0m0.607s
sys     0m12.812s
```

Baseline под стресс тестом:

```
user@debian:~/lab1/readwrite$ time ./graph_traverse --write 1 graph-rand.bin
time ./graph_traverse --write 1 graph-seq.bin
Iteration 1/1 (write): traversing graph-rand.bin ... OK (436905 nodes processed)

real    0m0.714s
user    0m0.164s
sys     0m0.550s
Iteration 1/1 (write): traversing graph-seq.bin ... OK (436905 nodes processed)

real    0m0.663s
user    0m0.152s
sys     0m0.511s
user@debian:~/lab1/readwrite$ time ./graph_traverse --write 1 graph-rand.bin
time ./graph_traverse --write 1 graph-seq.bin
Iteration 1/1 (write): traversing graph-rand.bin ... OK (436905 nodes processed)

real    0m1.043s
user    0m0.224s
sys     0m0.797s
Iteration 1/1 (write): traversing graph-seq.bin ... OK (436905 nodes processed)

real    0m0.967s
user    0m0.216s
sys     0m0.752s
```

Большинство вариантов работает достаточно быстро, больше 30 секунд работают только write no-cache. Т.к. исследовать no-cache мне не нужно, размер графа я оставлю неизменным.

```
Iteration 1/1 (read): traversing ../graph-rand.bin ... OK (436905 nodes processed)
        Command being timed: "./graph_traverse 1 ../graph-rand.bin"
        User time (seconds): 0.24
        System time (seconds): 0.63
        Percent of CPU this job got: 78%
        Elapsed (wall clock) time (h:mm:ss or m:ss): 0:01.12
        Average shared text size (kbytes): 0
        Average unshared data size (kbytes): 0
        Average stack size (kbytes): 0
        Average total size (kbytes): 0
        Maximum resident set size (kbytes): 1352
        Average resident set size (kbytes): 0
        Major (requiring I/O) page faults: 1
        Minor (reclaiming a frame) page faults: 78
        Voluntary context switches: 1703
        Involuntary context switches: 7
        Swaps: 0
        File system inputs: 20608
        File system outputs: 0
        Socket messages sent: 0
        Socket messages received: 0
        Signals delivered: 0
        Page size (bytes): 4096
        Exit status: 0
Iteration 1/1 (read): traversing ../graph-seq.bin ... OK (436905 nodes processed)
        Command being timed: "./graph_traverse 1 ../graph-seq.bin"
        User time (seconds): 0.20
        System time (seconds): 0.55
        Percent of CPU this job got: 99%
        Elapsed (wall clock) time (h:mm:ss or m:ss): 0:00.76
```

```
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 1344
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 1
Minor (reclaiming a frame) page faults: 81
Voluntary context switches: 7
Involuntary context switches: 8
Swaps: 0
File system inputs: 20608
File system outputs: 0
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0
```

4.3. Гипотеза

При записи программа займёт больше real time, чем при только чтении из-за большего числа переключений контекста.

4.4. План эксперимента

Для расчета числа запусков проводится с помощью обратной формулы доверительного интервала:

$N \approx (t \times \frac{s}{E})^2$

На пилотных 10 запусках получилось максимальное n=12, округлим до десятков в большую сторону.

1. Метрики: real time
2. Т.к. программа будет запускаться без флага `--no-cache`, то кэш должен участвовать в эксперименте. Будем его прогревать предварительным запуском экземпляров программы: один запуск read, второй write.
3. Программу будем запускать 20 раз. Будем чередовать запуски read и write.

4.5. Окружение перед экспериментом

```
user@debian:~$ top -bn1 | head -20
top - 13:20:03 up 2 min,  2 users,  load average: 0.06, 0.09, 0.03
Tasks: 140 total,   1 running, 139 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  2.4 sy,  0.0 ni, 97.6 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3921.3 total,   3693.1 free,    303.8 used,     95.8 buff/cache
MiB Swap:   1101.0 total,   1101.0 free,      0.0 used.   3617.4 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   23536  14172  10648 S   0.0   0.4   0:01.53 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      9 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker+
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     11 root      20   0       0      0      0 I   0.0   0.0   0:00.01 kworker+
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.19 kworker+
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
```

4.6. Измерения

Для измерения параметров был подготовлен bash скрипт, сохраняющий результат в csv файл:

```
#!/usr/bin/env bash
set -euo pipefail

bash ./warmup.sh

N=20
OUT=results.csv

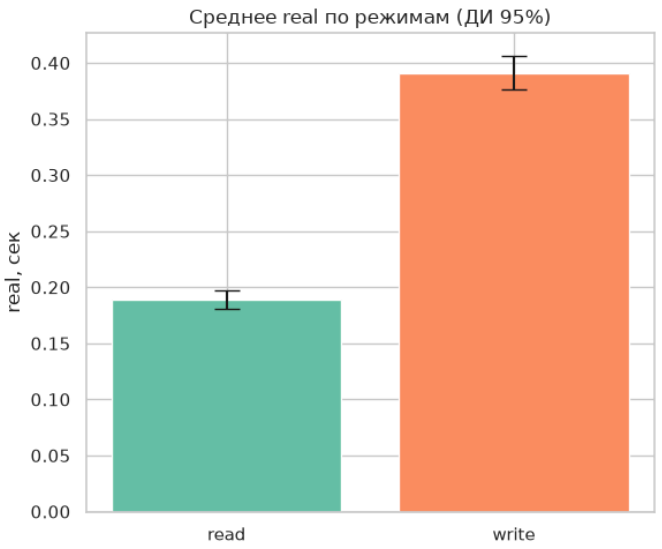
echo "n,mode,timestamp,real,user,sys" > "$OUT"

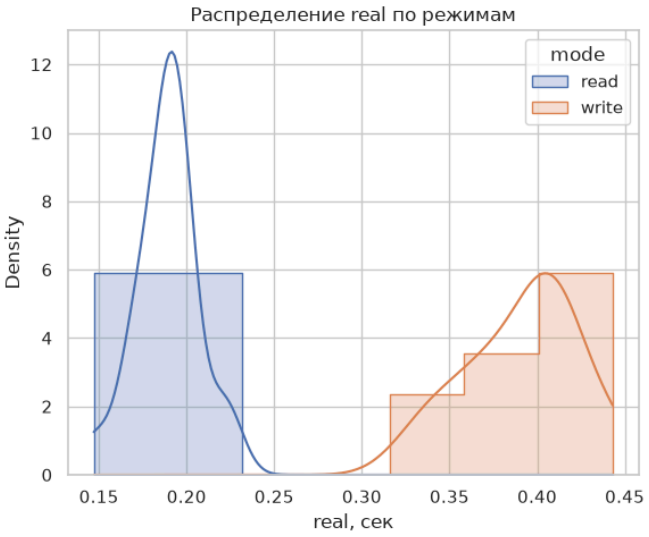
for i in $(seq 1 "$N"); do
  nice -n -20 taskset -c 0 /usr/bin/time -f "$i,read,$(date +%s),%e,%U,%S" -a -o "$OUT"
./graph_traverse 1 graph-seq.bin 2> /dev/null
  nice -n -20 taskset -c 0 /usr/bin/time -f "$i,write,$(date +%s),%e,%U,%S" -a -o "$OUT"
./graph_traverse --write 1 graph-seq.bin 2> /dev/null
done
```

4.7. Анализ измерений

В ходе анализа получены следующие результаты:

mode	N	mean	std	ci_low	ci_high	ci_halfwidth
read	20	0.1889	0.017335	0.180787	0.197013	0.008113
write	20	0.3913	0.031784	0.376425	0.406175	0.014875





4.8. Выводы

Полученные значения подтверждают гипотезу, read в среднем работает быстрее write. Результаты измерений были проанализированы, были построены графики: столбчатая диаграмма средних с усами ДИ и график распределения.

5. Исследование graph_traverse_mmap

5.1. Паспорт системы

Такой же, как и в исследовании graph_traverse.

5.2. Baseline

```
Iteration 1/1 (read): traversing graph-rand.bin ... OK (436905 nodes processed)

real    0m0.065s
user    0m0.065s
sys     0m0.000s
Iteration 1/1 (read): traversing graph-seq.bin ... OK (436905 nodes processed)

real    0m0.005s
user    0m0.000s
sys     0m0.005s
Iteration 1/1 (read): traversing graph-rand.bin ... OK (436905 nodes processed)

real    0m0.049s
user    0m0.050s
sys     0m0.000s
Iteration 1/1 (read): traversing graph-seq.bin ... OK (436905 nodes processed)

real    0m0.005s
user    0m0.005s
sys     0m0.000s
Iteration 1/1 (write): traversing graph-rand.bin ... OK (436905 nodes processed)
```

```
real    0m0.052s
user    0m0.052s
sys     0m0.000s
Iteration 1/1 (write): traversing graph-seq.bin ... OK (436905 nodes processed)

real    0m0.011s
user    0m0.000s
sys     0m0.011s
Iteration 1/1 (write): traversing graph-rand.bin ... OK (436905 nodes processed)

real    0m0.078s
user    0m0.061s
sys     0m0.008s
Iteration 1/1 (write): traversing graph-seq.bin ... OK (436905 nodes processed)

real    0m0.021s
user    0m0.003s
sys     0m0.009s
```

Baseline под стресс тестом (разница на уровне погрешности):

```
user@debian:~/lab1/mmap$ time ./graph_traverse --write 1 graph-rand.bin
time ./graph_traverse --write 1 graph-seq.bin
-bash: ./graph_traverse: No such file or directory

real    0m0.001s
user    0m0.001s
sys     0m0.000s
-bash: ./graph_traverse: No such file or directory

real    0m0.001s
user    0m0.000s
sys     0m0.000s
user@debian:~/lab1/mmap$ time ./graph_traverse --write 1 graph-rand.bin
time ./graph_traverse --write 1 graph-seq.bin
-bash: ./graph_traverse: No such file or directory

real    0m0.001s
user    0m0.000s
sys     0m0.000s
-bash: ./graph_traverse: No such file or directory

real    0m0.005s
user    0m0.000s
sys     0m0.000s
```

Большинство запусков программы отрабатывают слишком быстро. Для лучшей наглядности анализа увеличим размер графа до 100M.

```
Iteration 1/1 (read): traversing ../graph-rand.bin ... OK (436905 nodes processed)
        Command being timed: "./graph_traverse_mmap 1 ../graph-rand.bin"
        User time (seconds): 0.04
        System time (seconds): 0.01
        Percent of CPU this job got: 57%
        Elapsed (wall clock) time (h:mm:ss or m:ss): 0:00.09
        Average shared text size (kbytes): 0
        Average unshared data size (kbytes): 0
        Average stack size (kbytes): 0
        Average total size (kbytes): 0
        Maximum resident set size (kbytes): 11584
        Average resident set size (kbytes): 0
        Major (requiring I/O) page faults: 108
        Minor (reclaiming a frame) page faults: 243
        Voluntary context switches: 115
        Involuntary context switches: 0
        Swaps: 0
        File system inputs: 20592
        File system outputs: 0
```

```
        Socket messages sent: 0
        Socket messages received: 0
        Signals delivered: 0
        Page size (bytes): 4096
        Exit status: 0
Iteration 1/1 (read): traversing ../graph-seq.bin ... OK (436905 nodes processed)
        Command being timed: "./graph_traverse_mmap 1 ../graph-seq.bin"
        User time (seconds): 0.00
        System time (seconds): 0.01
        Percent of CPU this job got: 58%
        Elapsed (wall clock) time (h:mm:ss or m:ss): 0:00.03
        Average shared text size (kbytes): 0
        Average unshared data size (kbytes): 0
        Average stack size (kbytes): 0
        Average total size (kbytes): 0
        Maximum resident set size (kbytes): 11560
        Average resident set size (kbytes): 0
        Major (requiring I/O) page faults: 1
        Minor (reclaiming a frame) page faults: 241
        Voluntary context switches: 77
        Involuntary context switches: 0
        Swaps: 0
        File system inputs: 20600
        File system outputs: 0
        Socket messages sent: 0
        Socket messages received: 0
        Signals delivered: 0
        Page size (bytes): 4096
        Exit status: 0
```

5.3. Гипотеза

При записи программа займёт больше real time, чем при только чтении, но эта разница будет менее выраженной, чем при исследовании graph_traverse из-за отсутствия переключения контекста.

5.4. План эксперимента

Расчет N происходит так же как и в первом эксперименте.

1. Метрики: real time
2. Будем его прогревать предварительным запуском экземпляров программы: один запуск read, второй write.
3. Программу будем запускать 30 раз. Будем чередовать запуски read и write.

5.5. Окружение перед экспериментом

```
user@debian:~$ top -bn1 | head -20
top - 13:20:03 up 2 min,  2 users,  load average: 0.06, 0.09, 0.03
Tasks: 140 total,   1 running, 139 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  2.4 sy,  0.0 ni, 97.6 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3921.3 total,   3693.1 free,    303.8 used,     95.8 buff/cache
MiB Swap:   1101.0 total,   1101.0 free,      0.0 used.   3617.4 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   23536  14172  10648 S   0.0   0.4   0:01.53 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      9 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker+
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     11 root      20   0       0      0      0 I   0.0   0.0   0:00.01 kworker+
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.19 kworker+
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
```

5.6. Измерения

Для измерения параметров был подготовлен bash скрипт, сохраняющий результат в csv файл:

```
#!/usr/bin/env bash
set -euo pipefail

TIMEFORMAT="%3R,%3U,%3S"

bash ./warmup.sh

N=30
OUT=results.csv

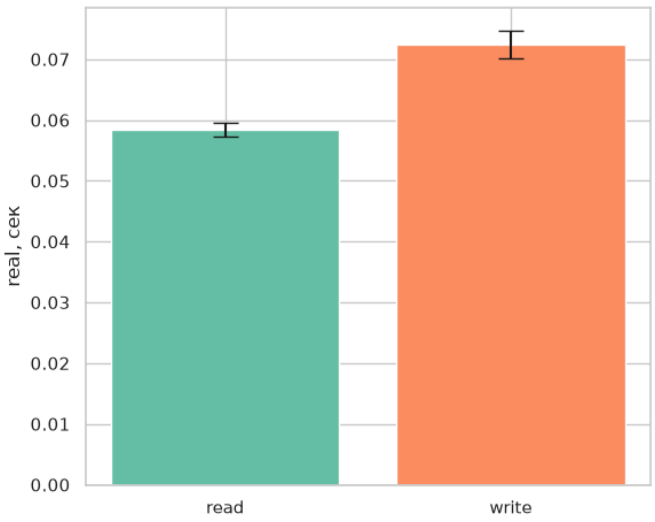
echo "n,mode,timestamp,real,user,sys" > "$OUT"

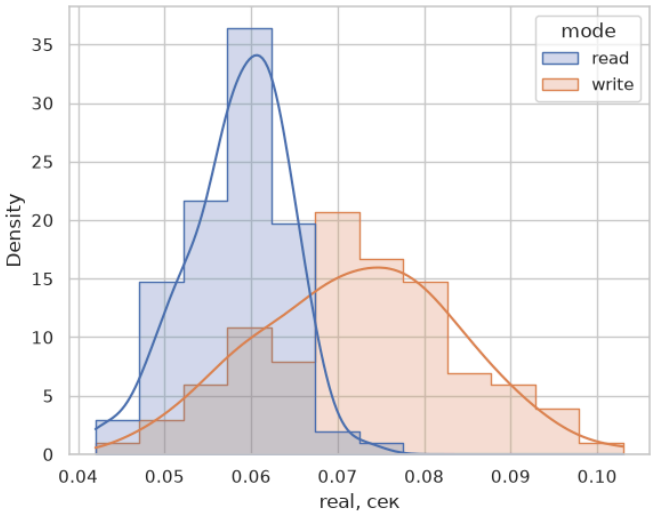
for i in $(seq 1 "$N"); do
  echo -n "$i,read," >> "$OUT"
  { time nice -n -20 taskset -c 0 ./graph_traverse_mmap 1 graph-seq.bin 2> /dev/null ; }
2>> "$OUT"
  echo -n "$i,write," >> "$OUT"
  { time nice -n -20 taskset -c 0 ./graph_traverse_mmap --write 1 graph-seq.bin 2>
/dev/null ; } 2>> "$OUT"
done
```

5.7. Анализ измерений

В ходе анализа получены следующие результаты:

mode	N	mean	std	ci_low	ci_high	ci_halfwidth
read	30	0.065800	0.007590	0.062966	0.068634	0.002834
write	30	0.081833	0.015102	0.076194	0.087473	0.005639





5.8. Выводы

Полученные значения подтверждают гипотезу, read в среднем работает быстрее write. Разница между временем read и write действительно менее выражена, чем в первом эксперименте. Результаты измерений были проанализированы, были построены графики: столбчатая диаграмма средних с усами ДИ и график распределения.

6. Вывод

Я провёл анализ работы двух программ, использующих файл, сравнил время их работы в разных режимах, проанализировал полученные результаты, построил графики на их основе. В ходе работы над заданием я узнал как правильно фиксировать окружение, выбирать инструменты, планировать число измерений.

```
>     uname -a
Linux debian 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29)
x86_64 GNU/Linux

     --------------

>     lscpu
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:             48 bits physical, 48 bits virtual
  Byte Order:                Little Endian
CPU(s):                      4
  On-line CPU(s) list:       0-3
Vendor ID:                   AuthenticAMD
  Model name:                AMD Ryzen 7 5825U with Radeon Graphics
    CPU family:              25
    Model:                   80
    Thread(s) per core:      1
    Core(s) per socket:      4
    Socket(s):               1
    Stepping:                0
    BogoMIPS:                3992.50
    Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdts
                             cp lm rep_good nopl xtopology cpuid extd_apicid tsc_known_freq
pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_t
                             imer aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy
svm cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw perfctr_core ssbd ibrs
                             ibpb stibp vmmcall fsgsbase tsc_adjust bmi1 avx2 smep bmi2
erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsave
                             s user_shstk clzero xsaveerptr wbnoinvd arat npt lbrv
nrip_save tsc_scale vmcb_clean flushbyasid pausefilter pfthreshold v_vmsave_vmload vgi
                             f umip pku ospke vaes vpclmulqdq rdpid overflow_recov succor
fsrm
Virtualization features:
  Virtualization:            AMD-V
  Hypervisor vendor:         KVM
  Virtualization type:       full
Caches (sum of all):
  L1d:                       256 KiB (4 instances)
  L1i:                       256 KiB (4 instances)
  L2:                        2 MiB (4 instances)
  L3:                        64 MiB (4 instances)
NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):         0-3
Vulnerabilities:
  Gather data sampling:      Not affected
  Indirect target selection: Not affected
  Itlb multihit:             Not affected
  L1tf:                      Not affected
  Mds:                       Not affected
  Meltdown:                  Not affected
  Mmio stale data:           Not affected
  Reg file data sampling:    Not affected
  Retbleed:                  Not affected
  Spec rstack overflow:      Mitigation; Safe RET
  Spec store bypass:         Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:                Mitigation; usercopy/swapgs barriers and __user pointer
sanitization
  Spectre v2:                Mitigation; Retpolines; IBPB conditional; IBRS_FW; STIBP
disabled; RSB filling; PBRSB-eIBRS Not affected; BHI Not affected
  Srbds:                     Not affected
  Tsa:                       Not affected
```

```
  Tsx async abort:           Not affected
  Vmscape:                   Not affected

     --------------

>     lscpu -e
CPU NODE SOCKET CORE L1d:L1i:L2:L3 ONLINE
  0    0      0    0 0:0:0:0          yes
  1    0      0    1 1:1:1:0          yes
  2    0      0    2 2:2:2:0          yes
  3    0      0    3 3:3:3:0          yes

     --------------

>     free -h
               total        used        free      shared  buff/cache   available
Mem:           3.8Gi       340Mi       3.5Gi       4.5Mi       202Mi       3.5Gi
Swap:          1.1Gi          0B       1.1Gi

     --------------

>     lshw
debian
    description: Computer
    product: Standard PC (Q35 + ICH9, 2009)
    vendor: QEMU
    version: pc-q35-11.1
    width: 64 bits
    capabilities: smbios-2.8 dmi-2.8 smp vsyscall32
    configuration: boot=normal uuid=097f7e80-b9a6-44aa-953c-d02e4b873de0
  *-core
       description: Motherboard
       physical id: 0
     *-firmware
          description: BIOS
          vendor: SeaBIOS
          physical id: 0
          version: Arch Linux 1.17.0-2-2
          date: 04/01/2014
          size: 96KiB
     *-cpu
          description: CPU
          product: AMD Ryzen 7 5825U with Radeon Graphics
          vendor: Advanced Micro Devices [AMD]
          physical id: 400
          bus info: cpu@0
          version: 25.80.0
          slot: CPU 0
          size: 2GHz
          capacity: 2GHz
          width: 64 bits
          capabilities: fpu fpu_exception wp vme de pse tsc msr pae mce cx8 apic sep mtrr
pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb
rdtscp x86-64 rep_good nopl xtopology cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3
fma cx16 sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand
hypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw
perfctr_core ssbd ibrs ibpb stibp vmmcall fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms
invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves user_shstk
clzero xsaveerptr wbnoinvd arat npt lbrv nrip_save tsc_scale vmcb_clean flushbyasid
pausefilter pfthreshold v_vmsave_vmload vgif umip pku ospke vaes vpclmulqdq rdpid
overflow_recov succor fsrm
          configuration: cores=4 enabledcores=4 microcode=173015058 threads=4
     *-memory
          description: System Memory
          physical id: 1000
          size: 4GiB
          capabilities: ecc
```

```
          configuration: errordetection=multi-bit-ecc
        *-bank
             description: DIMM RAM
             vendor: QEMU
             physical id: 0
             slot: DIMM 0
             size: 4GiB
     *-pci
          description: Host bridge
          product: 82G33/G31/P35/P31 Express DRAM Controller
          vendor: Intel Corporation
          physical id: 100
          bus info: pci@0000:00:00.0
          version: 00
          width: 32 bits
          clock: 33MHz
        *-display
             description: VGA compatible controller
             product: Virtio 1.0 GPU
             vendor: Red Hat, Inc.
             physical id: 1
             bus info: pci@0000:00:01.0
             logical name: /dev/fb0
             version: 01
             width: 64 bits
             clock: 33MHz
             capabilities: msix vga_controller bus_master cap_list rom fb
             configuration: depth=32 driver=virtio-pci latency=0 mode=1280x800
resolution=1280,800 visual=truecolor xres=1280 yres=800
             resources: iomemory:e0c0-e0bf irq:21 memory:fd000000-fd7fffff
memory:e0c00000000-e0c00003fff memory:fea14000-fea14fff memory:c0000-dffff
           *-virtio0 UNCLAIMED
                description: Virtual I/O device
                physical id: 0
                bus info: virtio@0
                configuration: driver=virtio_gpu
        *-pci:0
             description: PCI bridge
             product: QEMU PCIe Root port
             vendor: Red Hat, Inc.
             physical id: 2
             bus info: pci@0000:00:02.0
             version: 00
             width: 32 bits
             clock: 33MHz
             capabilities: pci pciexpress msix normal_decode bus_master cap_list
             configuration: driver=pcieport
             resources: irq:22 memory:fea15000-fea15fff ioport:1000(size=4096)
memory:fe800000-fe9fffff ioport:e0a00000000(size=8589934592)
           *-network
                description: Ethernet controller
                product: Virtio 1.0 network device
                vendor: Red Hat, Inc.
                physical id: 0
                bus info: pci@0000:01:00.0
                version: 01
                width: 64 bits
                clock: 33MHz
                capabilities: msix pm pciexpress bus_master cap_list rom
                configuration: driver=virtio-pci latency=0
                resources: iomemory:e0a0-e09f irq:22 memory:fe840000-fe840fff
memory:e0a00000000-e0a00003fff memory:fe800000-fe83ffff
              *-virtio1
                   description: Ethernet interface
                   physical id: 0
                   bus info: virtio@1
                   logical name: enp1s0
```

```
                  serial: 52:54:00:fb:68:51
                  capabilities: ethernet physical
                  configuration: autonegotiation=off broadcast=yes driver=virtio_net
driverversion=1.0.0 ip=192.168.0.108 link=yes multicast=yes
        *-pci:1
             description: PCI bridge
             product: QEMU PCIe Root port
             vendor: Red Hat, Inc.
             physical id: 2.1
             bus info: pci@0000:00:02.1
             version: 00
             width: 32 bits
             clock: 33MHz
             capabilities: pci pciexpress msix normal_decode bus_master cap_list
             configuration: driver=pcieport
             resources: irq:22 memory:fea16000-fea16fff ioport:2000(size=4096)
memory:fe600000-fe7fffff ioport:e0800000000(size=8589934592)
           *-usb
                description: USB controller
                product: QEMU XHCI Host Controller
                vendor: Red Hat, Inc.
                physical id: 0
                bus info: pci@0000:02:00.0
                version: 01
                width: 64 bits
                clock: 33MHz
                capabilities: msix pciexpress xhci bus_master cap_list
                configuration: driver=xhci_hcd latency=0
                resources: irq:22 memory:fe600000-fe603fff
              *-usbhost:0
                   product: xHCI Host Controller
                   vendor: Linux 6.12.107+deb13-amd64 xhci-hcd
                   physical id: 0
                   bus info: usb@1
                   logical name: usb1
                   version: 6.12
                   capabilities: usb-2.00
                   configuration: driver=hub slots=15 speed=480Mbit/s
                 *-usb:0 UNCLAIMED
                      description: Smart card reader
                      product: QEMU USB CCID
                      vendor: QEMU
                      physical id: 1
                      bus info: usb@1:1
                      version: 0.00
                      serial: 1-0000:00:02.1:00.0-1
                      capabilities: usb-1.10
                      configuration: maxpower=100mA speed=12Mbit/s
                 *-usb:1
                      description: Human interface device
                      product: QEMU QEMU USB Tablet
                      vendor: QEMU
                      physical id: 2
                      bus info: usb@1:2
                      logical name: input4
                      logical name: /dev/input/event3
                      logical name: /dev/input/js0
                      logical name: /dev/input/mouse2
                      version: 0.00
                      serial: 28754-0000:00:02.1:00.0-2
                      capabilities: usb-2.00 usb
                      configuration: driver=usbhid maxpower=100mA speed=480Mbit/s
              *-usbhost:1
                   product: xHCI Host Controller
                   vendor: Linux 6.12.107+deb13-amd64 xhci-hcd
                   physical id: 1
                   bus info: usb@2
```

```
                  logical name: usb2
                  version: 6.12
                  capabilities: usb-3.00
                  configuration: driver=hub slots=15 speed=5000Mbit/s
        *-pci:2
             description: PCI bridge
             product: QEMU PCIe Root port
             vendor: Red Hat, Inc.
             physical id: 2.2
             bus info: pci@0000:00:02.2
             version: 00
             width: 32 bits
             clock: 33MHz
             capabilities: pci pciexpress msix normal_decode bus_master cap_list
             configuration: driver=pcieport
             resources: irq:22 memory:fea17000-fea17fff ioport:3000(size=4096)
memory:fe400000-fe5fffff ioport:e0600000000(size=8589934592)
           *-communication
                description: Communication controller
                product: Virtio 1.0 console
                vendor: Red Hat, Inc.
                physical id: 0
                bus info: pci@0000:03:00.0
                version: 01
                width: 64 bits
                clock: 33MHz
                capabilities: msix pm pciexpress bus_master cap_list
                configuration: driver=virtio-pci latency=0
                resources: iomemory:e060-e05f irq:22 memory:fe400000-fe400fff
memory:e0600000000-e0600003fff
              *-virtio2 UNCLAIMED
                   description: Virtual I/O device
                   physical id: 0
                   bus info: virtio@2
                   configuration: driver=virtio_console
        *-pci:3
             description: PCI bridge
             product: QEMU PCIe Root port
             vendor: Red Hat, Inc.
             physical id: 2.3
             bus info: pci@0000:00:02.3
             version: 00
             width: 32 bits
             clock: 33MHz
             capabilities: pci pciexpress msix normal_decode bus_master cap_list
             configuration: driver=pcieport
             resources: irq:22 memory:fea18000-fea18fff ioport:4000(size=4096)
memory:fe200000-fe3fffff ioport:e0400000000(size=8589934592)
           *-scsi
                description: SCSI storage controller
                product: Virtio 1.0 block device
                vendor: Red Hat, Inc.
                physical id: 0
                bus info: pci@0000:04:00.0
                version: 01
                width: 64 bits
                clock: 33MHz
                capabilities: scsi msix pm pciexpress bus_master cap_list
                configuration: driver=virtio-pci latency=0
                resources: iomemory:e040-e03f irq:22 memory:fe200000-fe200fff
memory:e0400000000-e0400003fff
              *-virtio3
                   description: Virtual I/O device
                   physical id: 0
                   bus info: virtio@3
                   logical name: /dev/vda
                   size: 20GiB (21GB)
```

```
                  capabilities: partitioned partitioned:dos
                  configuration: driver=virtio_blk logicalsectorsize=512 sectorsize=512
signature=1f2c193a
               *-volume:0
                    description: EXT4 volume
                    vendor: Linux
                    physical id: 1
                    bus info: virtio@3,1
                    logical name: /dev/vda1
                    logical name: /
                    version: 1.0
                    serial: 51ceaec8-c5b4-420c-a634-208017f72332
                    size: 18GiB
                    capacity: 18GiB
                    capabilities: primary bootable journaled extended_attributes
large_files huge_files dir_nlink recover 64bit extents ext4 ext2 initialized
                    configuration: created=2026-09-27 19:56:58 filesystem=ext4
lastmountpoint=/ modified=2026-09-27 23:02:17 mount.fstype=ext4
mount.options=rw,relatime,errors=remount-ro mounted=2026-09-27 21:01:07 state=mounted
               *-volume:1
                    description: Extended partition
                    physical id: 2
                    bus info: virtio@3,2
                    logical name: /dev/vda2
                    size: 1101MiB
                    capacity: 1101MiB
                    capabilities: primary extended partitioned partitioned:extended
                  *-logicalvolume
                       description: Linux swap volume
                       physical id: 5
                       logical name: /dev/vda5
                       version: 1
                       serial: 015086bc-8dd4-4ee3-85d2-3da21cdf1cb5
                       size: 1101MiB
                       capacity: 1101MiB
                       capabilities: nofs swap initialized
                       configuration: filesystem=swap pagesize=4096
       *-pci:4
            description: PCI bridge
            product: QEMU PCIe Root port
            vendor: Red Hat, Inc.
            physical id: 2.4
            bus info: pci@0000:00:02.4
            version: 00
            width: 32 bits
            clock: 33MHz
            capabilities: pci pciexpress msix normal_decode bus_master cap_list
            configuration: driver=pcieport
            resources: irq:22 memory:fea19000-fea19fff ioport:5000(size=4096)
memory:fe000000-fe1fffff ioport:e0200000000(size=8589934592)
          *-generic
               description: Unclassified device
               product: Virtio 1.0 balloon
               vendor: Red Hat, Inc.
               physical id: 0
               bus info: pci@0000:05:00.0
               version: 01
               width: 64 bits
               clock: 33MHz
               capabilities: msix pm pciexpress bus_master cap_list
               configuration: driver=virtio-pci latency=0
               resources: iomemory:e020-e01f irq:22 memory:fe000000-fe000fff
memory:e0200000000-e0200003fff
             *-virtio4 UNCLAIMED
                  description: Virtual I/O device
                  physical id: 0
                  bus info: virtio@4
```

```
                configuration: driver=virtio_balloon
        *-pci:5
             description: PCI bridge
             product: QEMU PCIe Root port
             vendor: Red Hat, Inc.
             physical id: 2.5
             bus info: pci@0000:00:02.5
             version: 00
             width: 32 bits
             clock: 33MHz
             capabilities: pci pciexpress msix normal_decode bus_master cap_list
             configuration: driver=pcieport
             resources: irq:22 memory:fea1a000-fea1afff ioport:6000(size=4096)
memory:fde00000-fdffffff ioport:e0000000000(size=8589934592)
        *-multimedia
             description: Audio device
             product: 82801I (ICH9 Family) HD Audio Controller
             vendor: Intel Corporation
             physical id: 1b
             bus info: pci@0000:00:1b.0
             logical name: card0
             logical name: /dev/snd/controlC0
             logical name: /dev/snd/hwC0D0
             logical name: /dev/snd/pcmC0D0c
             logical name: /dev/snd/pcmC0D0p
             version: 03
             width: 32 bits
             clock: 33MHz
             capabilities: msi bus_master cap_list
             configuration: driver=snd_hda_intel latency=0
             resources: irq:51 memory:fea10000-fea13fff
        *-isa
             description: ISA bridge
             product: 82801IB (ICH9) LPC Interface Controller
             vendor: Intel Corporation
             physical id: 1f
             bus info: pci@0000:00:1f.0
             version: 02
             width: 32 bits
             clock: 33MHz
             capabilities: isa
             configuration: driver=lpc_ich latency=0
             resources: irq:0
           *-pnp00:00
                product: PnP device PNP0501
                physical id: 0
                capabilities: pnp
                configuration: driver=serial
           *-pnp00:01
                product: PnP device PNP0303
                physical id: 1
                capabilities: pnp
                configuration: driver=i8042 kbd
           *-pnp00:02
                product: PnP device PNP0f13
                physical id: 2
                capabilities: pnp
                configuration: driver=i8042 aux
           *-pnp00:03
                product: PnP device PNP0b00
                physical id: 3
                capabilities: pnp
                configuration: driver=rtc_cmos
           *-pnp00:04
                product: PnP device PNP0c01
                physical id: 4
                capabilities: pnp
```

```
               configuration: driver=system
       *-sata
            description: SATA controller
            product: 82801IR/IO/IH (ICH9R/DO/DH) 6 port SATA Controller [AHCI mode]
            vendor: Intel Corporation
            physical id: 1f.2
            bus info: pci@0000:00:1f.2
            logical name: scsi2
            version: 02
            width: 32 bits
            clock: 33MHz
            capabilities: sata msi ahci_1.0 bus_master cap_list emulated
            configuration: driver=ahci latency=0
            resources: irq:42 ioport:c040(size=32) memory:fea1b000-fea1bfff
          *-cdrom
               description: DVD reader
               product: QEMU DVD-ROM
               vendor: QEMU
               physical id: 0.0.0
               bus info: scsi@2:0.0.0
               logical name: /dev/cdrom
               logical name: /dev/sr0
               version: 2.5+
               capabilities: removable audio dvd
               configuration: ansiversion=5 status=nodisc
       *-serial
            description: SMBus
            product: 82801I (ICH9 Family) SMBus Controller
            vendor: Intel Corporation
            physical id: 1f.3
            bus info: pci@0000:00:1f.3
            version: 02
            width: 32 bits
            clock: 33MHz
            configuration: driver=i801_smbus latency=0
            resources: irq:16 ioport:700(size=64)
  *-input:0
       product: AT Translated Set 2 keyboard
       physical id: 1
       logical name: input0
       logical name: /dev/input/event0
       logical name: input0::capslock
       logical name: input0::numlock
       logical name: input0::scrolllock
       capabilities: i8042
  *-input:1
       product: VirtualPS/2 VMware VMMouse
       physical id: 2
       logical name: input2
       logical name: /dev/input/event2
       logical name: /dev/input/mouse1
       capabilities: i8042
  *-input:2
       product: VirtualPS/2 VMware VMMouse
       physical id: 3
       logical name: input3
       logical name: /dev/input/event1
       logical name: /dev/input/mouse0
       capabilities: i8042
  *-input:3
       product: Power Button
       physical id: 4
       logical name: input5
       logical name: /dev/input/event4
       capabilities: platform
  *-input:4
       product: PC Speaker
```

```
        physical id: 5
        logical name: input6
        logical name: /dev/input/event5
        capabilities: isa

      ---------------

>      nproc
4
```